

EDWIN CALLES MOLINA

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U.S. Citizen — Eligible for U.S. Government Security Clearance

OBJECTIVE

Aerospace Engineering student (GPA 3.97) seeking a Summer 2027 engineering internship in aerospace structures, propulsion, or manufacturing.

SUMMARY

Hands-on rocketry experience leading sub-teams through design, 3D-printed prototyping, avionics integration, and flight testing — including a NAR Level 2 certification flight and a supersonic flight test whose recovered data drove an airframe redesign.

EDUCATION

California State Polytechnic University, Pomona

Pomona, CA

B.S. Aerospace Engineering

Expected Graduation: August 2029

GPA: 3.97 | **Honors:** President's Honor List (2025–2026)

Coursework: Calculus I–III, Advanced Engineering Mathematics, Physics I (Mechanics), General Chemistry

In Progress (Fall 2026): Engineering Statics, Materials Science & Engineering, Aerospace Computational Methods, Fundamentals of Systems Engineering & Design, Engineering Graphics

EXPERIENCE & PROJECTS

Undergraduate Missiles, Ballistics, and Rocketry Association (UMBRA)

Pomona, CA

Team Lead, Electronics & Launch Operations — Project EXO Bronco

May 2026 – Present

- Co-lead the Electronics & Launch Operations sub-team for EXO Bronco, a two-stage rocket targeting the edge of space, responsible for the vehicle's avionics.
- Wrote the requirements and documentation structure the sub-team now works from, improving design tracking and communication between teams.
- Wired and integrated the avionics bay, completing pre-flight checkout and flight readiness preparation.

Sub-Team Lead, Small Rocket Competition (SRC)

Sept. 2025 – Dec. 2025

- Directed a 4-person team across research and design assignments.
- Designed the airframe and fin layout for a 2,800 ft target altitude, checking stability margin in OpenRocket simulation.
- Produced and reviewed 3D-printed prototype parts, revising the design based on fit and test results.
- Built and programmed an onboard system that recorded temperature and altitude in flight.

Independent Design Project — Dual-Deploy High Power Rockets

June 2026 – Present

- Designed, built, and flew two dual-deploy high power rockets across four flight tests, earning NAR Level 2 High Power certification on a 4,135 ft flight.
- Designed and 3D-printed the onboard electronics bays, which protected the recovery electronics through launch, ejection, and landing on every flight.
- Test-flew the second vehicle to 2,833 ft, then flew it supersonic to Mach 1.2 (1,367 ft/s), where it broke apart in flight; recovered the electronics and extracted the full flight data.
- Analyzed the flight data and recovered wreckage to determine that fin failure at supersonic speed caused the breakup, then specified a stiffer fin and composite airframe redesign.

Particle Image Velocimetry (PIV) – Design Team Systems Liaison

August 2026 – Present

Gurney & Wheelchair Transport Technician — Transport Today (Contract)

June 2024 – Present

- Transport non-emergency patients by gurney and wheelchair between hospitals, clinics, and care facilities, following safety procedure through loading, transit, and unloading.
- Operate electric lift ramp equipment to load, secure, and unload patients without incident.

TECHNICAL SKILLS

CAD & Simulation: SolidWorks (part modeling, assemblies, drawings), OpenRocket flight simulation

Programming: C++, Java, embedded sensor and data-logging code

Manufacturing & Hardware: FDM 3D printing / additive manufacturing, soldering, wiring harness assembly

Analysis & Documentation: Flight data analysis, failure investigation, Microsoft Excel, technical reporting, requirements documentation, Gantt Chart Creation

Certifications: NAR Level 2 High Power Rocketry Certification (August 2026); NAR Level 1 Certification